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THE INFLUENCE OF THE X4 VARIABLE ON 3D GRAPHICS FOR VEU

The article below discusses the issue of calculating the variable X4, which affects the values of GDP (GDP). The values obtained during the calculations are shown in the form of 3D graphs for X4. In this case, the variables are constant, decrease and increase. That is, the article considers the dependence of the change $X4 = f(X1, X2, X3, X5, X6)$ [1, 2, 3].

Here, Figure 1 shows a 3D graph for X4 at $X1 = X3 = 1.01$, $X2 = X6 = 1.10$, $X5 = 10$, where the values of X4 have a minimum of 0.91109 at point 6. The following Figure 2 gives a visual representation that for variables $X1 = X6 = 1.10$, $X2 = X3 = 1.01$, $X5 = 10$ The 3D graph has two zero values at points 6 and 7.

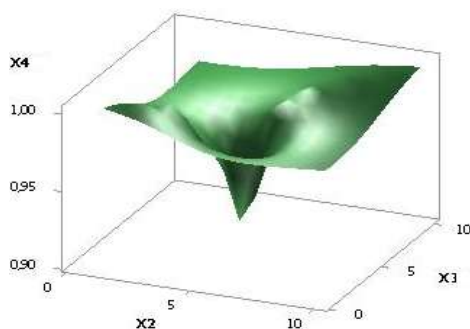


Рис. 1. $X4 = f(X1, X2, X3, X5, X6)$

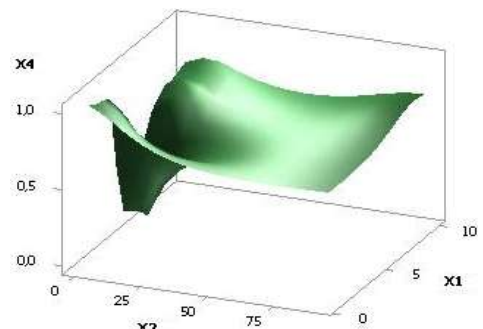


Рис. 2. $X4 = f(X1, X2, X3, X5, X6)$

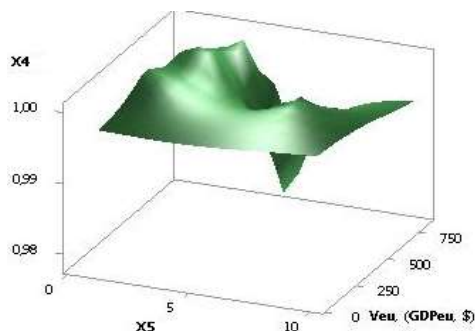


Рис. 3. $X4 = f(X1, X2, X3, X5, X6)$

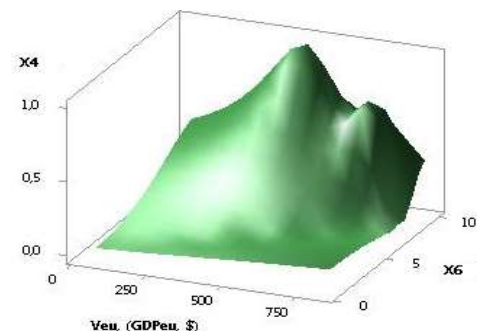


Рис. 4. $X4 = f(X1, X2, X3, X5, X6)$

The following two Figures 3 and 4 show 3D graphs for $X4 = f(X1, X2, X3, X5, X6)$ when the variables were $X1 = X2 = X6 = 1.10$, $X4 = 1.01$, $X5 = 10$ and $X1 = X2 = X4 = 1.01$, $X5 = 10$,

$X_6 = 1..10$ respectively. Here in Figure 3, the value of X_4 has a minimum of 0.978529 at point 7, and in Figure 4, the value of X_4 increases by 1.28 times.

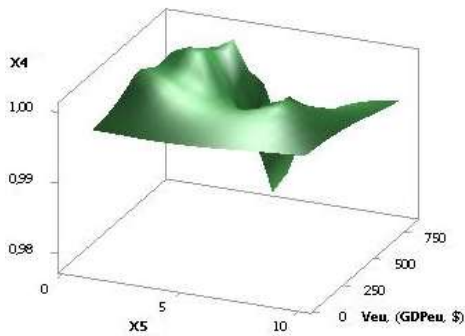


Рис. 5. $X_4 = f(X_1, X_2, X_3, X_5, X_6)$

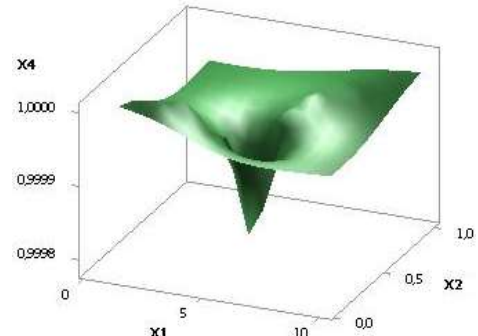


Рис. 6. $X_4 = f(X_1, X_2, X_3, X_5, X_6)$

The following two figures 5 and 6 were constructed at $X_1 = X_3 = 1..01$, $X_2 = 1..10$, $X_5 = 10$, $X_6 = 10..1$ and $X_1 = 1..10$, $X_2 = X_3 = 1..01$, $X_5 = 10$, $X_6 = 10..1$. Here in Figures 5 and 6, the constructed 3D graphs for X_4 have minima of 0.9785 and 0.999787 at points 6.

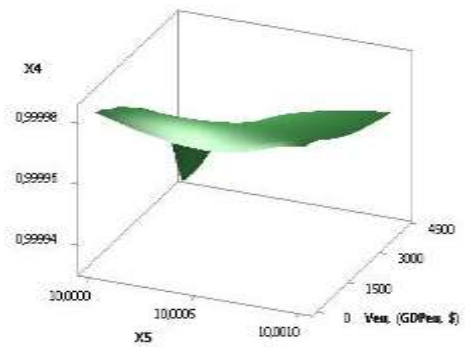


Рис. 7. $X_4 = f(X_1, X_2, X_3, X_5, X_6)$

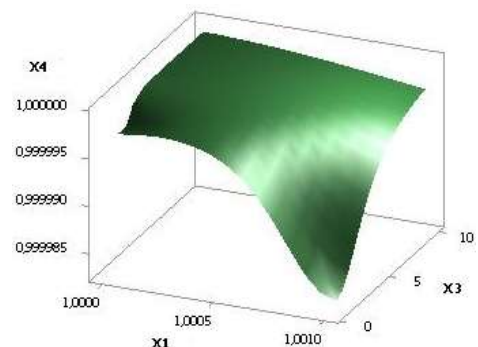


Рис. 8. $X_4 = f(X_1, X_2, X_3, X_5, X_6)$

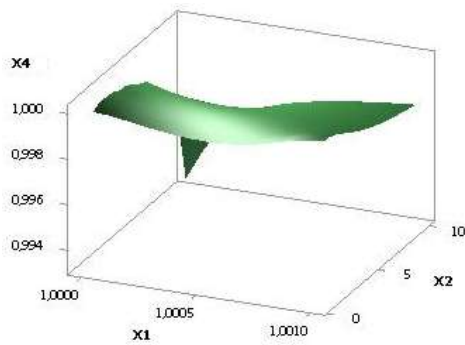


Рис. 9. $X_4 = f(X_1, X_2, X_3, X_5, X_6)$

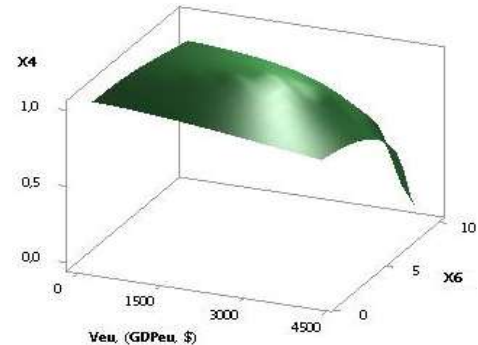


Рис. 10. $X_4 = f(X_1, X_2, X_3, X_5, X_6)$

The following values of variables $X_1 = X_2 = 1..10$, $X_3 = 1..01$, $X_5 = 10$, $X_6 = 10..1$ and $X_1 = X_2 = X_3 = 1..01$, $X_5 = 10$, $X_6 = 10$ were used to construct two 3D graphs in Figures 7 and 8..1. In Figure 7, the presented 3D graph for X_4 times

decreases slightly by 1,00005 times, and in Figure 8, the 3D graph for X_3 increases by 1,00002 times.

Figures 9 and 10 were constructed at $X_1=1..01$, $X_2=X_6=1..10$, $X_3=0.1..1$, $X_5=10$ and $X_1=X_6=1..10$, $X_2=1..01$, $X_3=0.1..1$, $X_5=10$. Here in Fig. 9, the values of X_4 decrease by 1,007 times, and in Fig. 10 they decrease by 1.62 times.

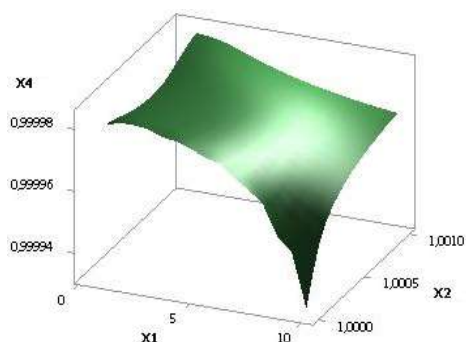


Рис. 11. $X_4 = f(X_1, X_2, X_3, X_5, X_6)$

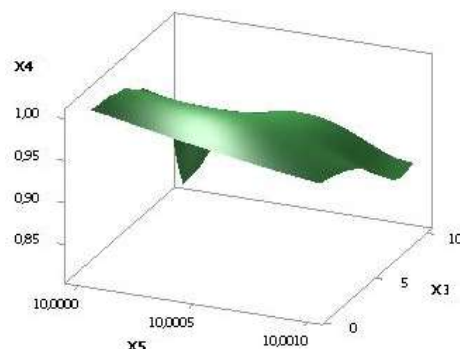


Рис. 12. $X_4 = f(X_1, X_2, X_3, X_5, X_6)$

The last figures 11 and 12 were constructed at $X_1=X_2=X_6=1..10$, $X_3=0.1..1$, $X_5=10$ and $X_1=X_2=10..1$, $X_3=0.1..1$, $X_5=10$, $X_6=1..10$. Here in Figure 11, the values of X_4 decrease 1,00005 times, and in Fig. 12 they increase by 1.23 times.

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